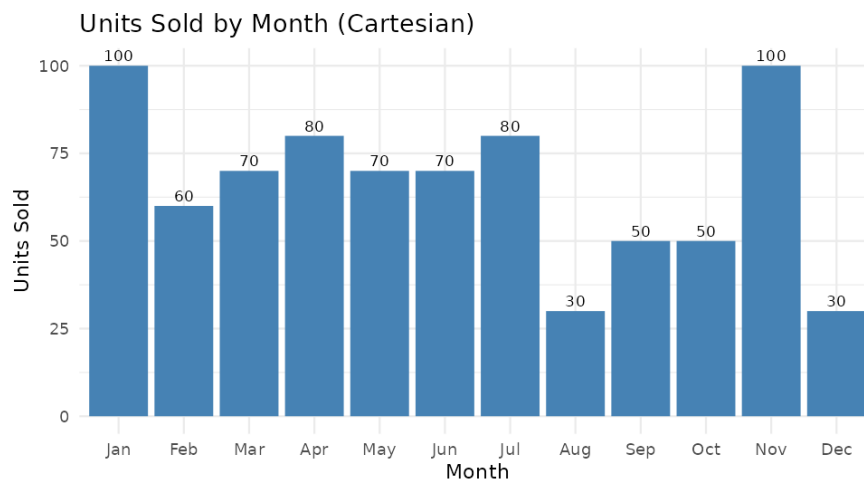


Q1. Table to Visual: Cartesian versus polar coordinates [10 marks]

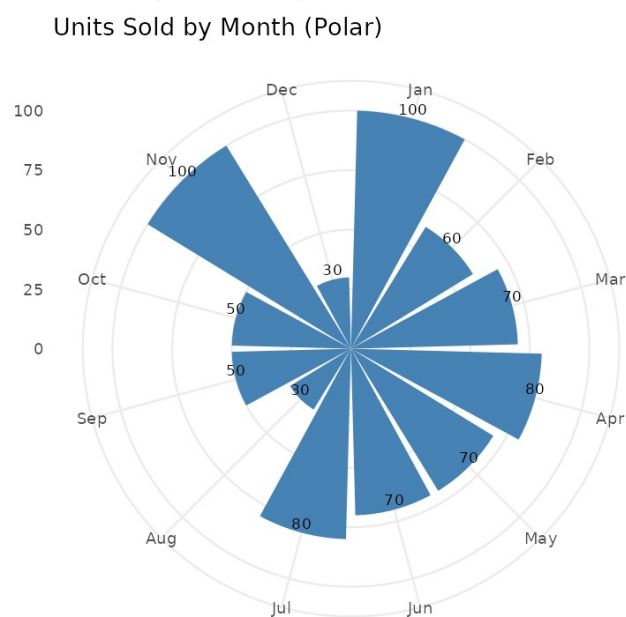
R program used to draw both charts

```
library(ggplot2)
m <- c("Jan", "Feb", "Mar", "Apr", "May", "Jun", "Jul", "Aug", "Sep", "Oct", "Nov", "Dec")
df <- data.frame(Month = factor(m, levels = m),
                 Units_Sold = c(100, 60, 70, 80, 70, 70, 80, 30, 50, 50, 100, 30))
# (a) Cartesian bar chart
p1 <- ggplot(df, aes(Month, Units_Sold)) +
  geom_col(fill = "steelblue") +
  geom_text(aes(label = Units_Sold), vjust = -0.4, size = 3) +
  labs(title = "Units Sold by Month (Cartesian)", x = "Month", y = "Units Sold") +
  theme_minimal()
# (b) Same data in polar coordinates (radial bars)
p2 <- p1 + coord_polar() + labs(title = "Units Sold by Month (Polar)", x = NULL, y = NULL)
print(p1); print(p2)
ggsave("q1_cart.png", p1, width = 6.5, height = 3.6, dpi = 150, bg = "white")
ggsave("q1_polar.png", p2, width = 5, height = 5, dpi = 150, bg = "white")
```

(a) Bar chart in Cartesian coordinates [3]



(b) Same data in polar coordinates (radial bars) [4]



Angle represents the Month (Jan to Dec around the circle).

Radius represents Units_Sold (a longer bar means more units).

(c) Better coordinate system [3]

Better: Cartesian, because bar heights on a straight axis are easiest to compare. For example, Jan and Nov (100) are clearly highest and Aug and Dec (30) lowest.

Polar advantage: It shows the yearly cycle, with Dec flowing back into Jan.

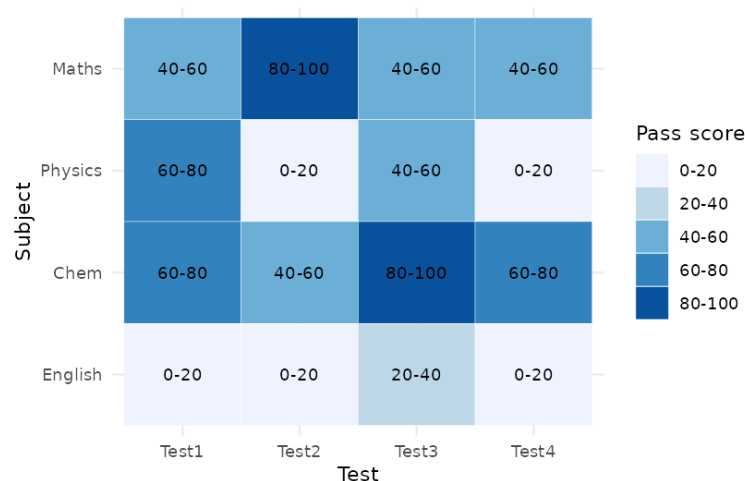
Polar drawback: Outer bars look bigger (area distortion), and lengths along a circle are hard to compare.

Q2. Visual to Table: Reading a colour scale [10 marks]

(a) Reconstructed 4 x 4 table (class intervals) [5]

Subject	Test1	Test2	Test3	Test4
Maths	40-60	80-100	40-60	40-60
Physics	60-80	0-20	40-60	0-20
Chem	60-80	40-60	80-100	60-80
English	0-20	0-20	20-40	0-20

Re-plotted in R from the table above (it matches the given heat map):



(b) Type of colour scale [3]

Sequential. The pass score is ordered from low (0) to high (100), so a single hue (blue) runs from light to dark. There is no meaningful midpoint, so a diverging scale is not needed, and the classes are ordered, so a qualitative scale would be wrong.

(c) Highest class [2]

80-100 class: Maths-Test2 and Chem-Test3.

Why darker = higher: More ink or depth naturally reads as "more", so readers link darker cells with higher scores without checking the legend.

Q3. Design Decision: Colour to distinguish and to highlight [10 marks]

(a) What is wrong with a different colour for every bar [3]

- Six bright colours compete equally, so nothing stands out and the missed-target bar is lost.
- Colours suggest the bars are different categories, which adds meaning that isn't there, and the reader must keep checking a legend.
- Many hues are hard for colour-blind readers to tell apart.

(b) Better colour scheme [4]

Highlight colour: Red or orange for the product line that missed its target. A warm, saturated colour signals a problem and grabs attention first.

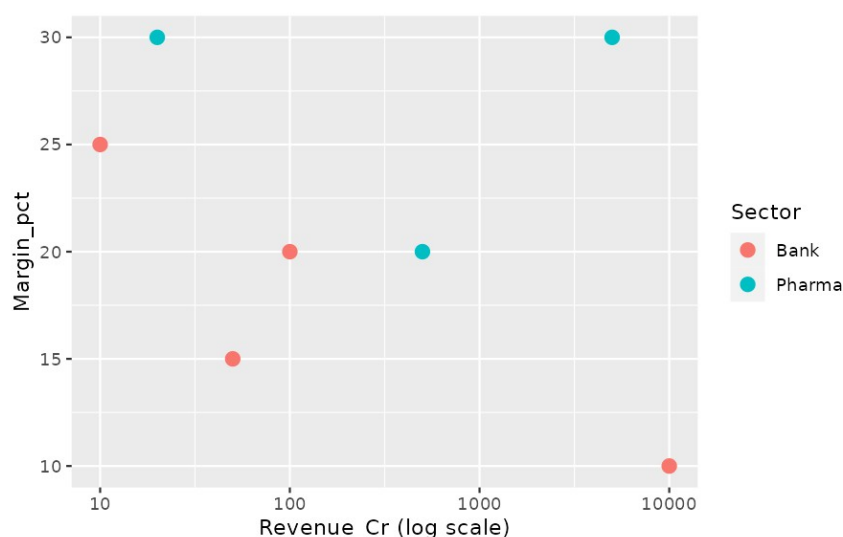
Other bars: Light grey (one neutral colour), so they act as background context and do not compete with the highlight.

(c) Non-colour cue [3]

- Add a dashed target line, and an annotation next to the highlighted bar such as "Missed target by X %".
- Sort the bars by value so the low performer sits at one end, and label the values directly on the bars.

Q4. R Code to Visual: Aesthetics and a log axis [10 marks]

(a) Chart produced by the program [5]



- x-axis (log10): ticks at 10, 100, 1000, 10000, labelled "Revenue_Cr (log scale)". y-axis: ticks at 10, 15, 20, 25, 30, labelled "Margin_pct".
- Points: Zeta (10, 25), Alpha (20, 30), Gamma (50, 15), Kappa (100, 20), Sigma (5000, 20), Delta (5000, 30), Beta (10000, 10).
- Legend "Sector": Bank in red/salmon (Gamma, Beta, Zeta, Kappa) and Pharma in teal (Delta, Sigma, Alpha), all points of size 3.

(b) Aesthetics, scale types and coordinate system [3]

Variable	Aesthetic	Scale type
Revenue_Cr	x position	Continuous, log10
Margin_pct	y position	Continuous, linear
Sector	colour	Discrete (nominal), default hue
Name	not mapped	—

Coordinate system: Cartesian (coord_cartesian, the default).

(c) Colour-blind-safe palette [2]

```
scale_x_log10() + scale_colour_viridis_d() +
```

Palette: viridis (discrete). Alternatives are scale_colour_brewer(palette = "Dark2") or the Okabe-Ito palette.



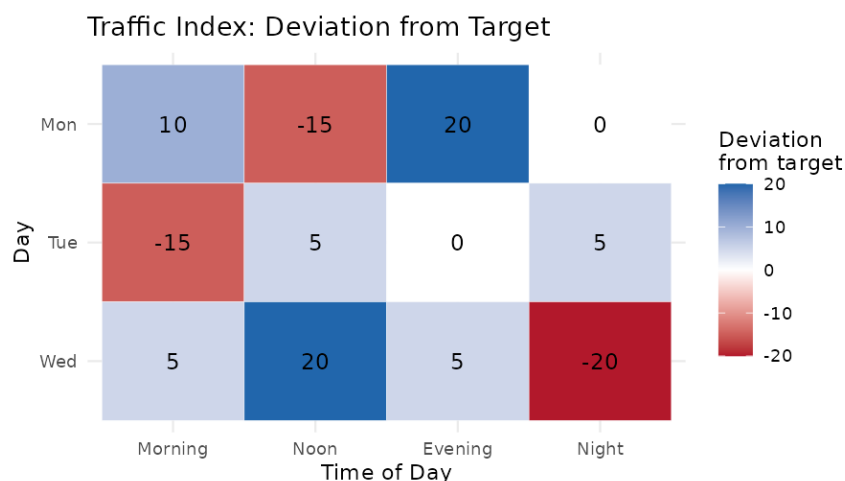
Q5. Visual to R Code: Choosing an effective colour scale [10 marks]

(a) Complete R program [7]

```
library(ggplot2)
# Data in long format: one row per Day x Time cell
df <- data.frame(
  Day = rep(c("Mon", "Tue", "Wed"), each = 4),
  Time = rep(c("Morning", "Noon", "Evening", "Night"), times = 3),
  Deviation = c( 10, -15, 20, 0, # Mon
                -15, 5, 0, 5, # Tue
                5, 20, 5, -20) # Wed
)
df$Day <- factor(df$Day, levels = c("Wed", "Tue", "Mon")) # Mon at top
df$Time <- factor(df$Time, levels = c("Morning", "Noon", "Evening", "Night"))

ggplot(df, aes(x = Time, y = Day, fill = Deviation)) +
  geom_tile(colour = "white") +
  geom_text(aes(label = Deviation), size = 4) +
  scale_fill_gradient2(low = "#B2182B", mid = "white", high = "#2166AC",
                      midpoint = 0, limits = c(-20, 20),
                      name = "Deviation\nfrom target") +
  labs(title = "Traffic Index: Deviation from Target",
       x = "Time of Day", y = "Day") +
  theme_minimal()
ggsave("q5_heat.png", width = 6, height = 3.4, dpi = 150, bg = "white")
```

Output



(b) Why a diverging scale with midpoint 0 [3]

- The data is a deviation from target, with both negative and positive values, and 0 means "on target".
- A diverging scale shows 0 as white, below-target values in red and above-target values in blue, with darkness showing the size of the deviation.

- A sequential scale would hide the sign (it treats -20 as simply "low"). A rainbow scale has no natural order or midpoint, creates false boundaries, and is not colour-blind-safe.